

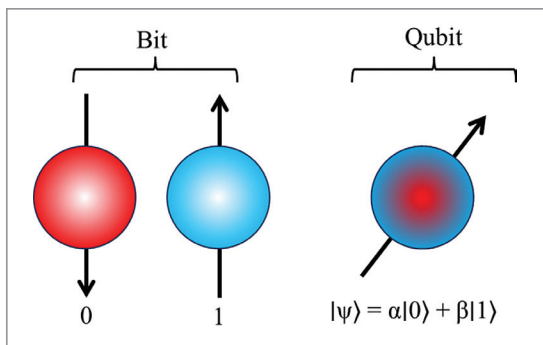


Fig. 2: Quantum superposition

states remain correlated, regardless of their distance. Imagine two dice that always show the same number, even if one is rolled on Earth and the other on Mars. This connection enables quantum teleportation and ultra-secure communication, as any attempt to measure or interfere with one particle instantly affects the other. Fig. 3 pictorially illustrates the concept of entanglement and how measurements of one entangled partner relate to those of the other.

No-cloning theorem. In classical computing, data can be copied endlessly, but in quantum mechanics, perfect duplication is impossible. This principle, known as the no-cloning theorem, ensures the security of quantum encryption. Any eavesdropper attempting to intercept quantum information will inevitably alter it, making their presence detectable. This property underpins quantum key distribution (QKD) and other secure communication protocols.

Uncertainty principle. In classical physics, we can measure an object's position and speed precisely. In contrast, quantum mechanics dictates that the more accurately we measure one, the less precisely we know the other. This is Heisenberg's Uncertainty Principle—a fundamental property of nature, not merely a limitation of instruments. Imagine a blurry photo of a moving car—if the image is sharp,

the position is clear, but the motion is not; if it is blurred, the motion is evident, but the exact location is lost. Similarly, in quantum systems, pinpointing one property disturbs the other. This principle has profound implications for precision measurement, quantum computing, and cryptography, shaping the future

of quantum technologies.

Coherence. Quantum systems rely on maintaining coherence, meaning the quantum states must remain in sync. Think of a choir singing in perfect harmony: as long as the singers stay coordinated, the music remains clear and beautiful. In quantum mechanics, coherence ensures that superposition and entanglement remain stable. However, just like noise can disrupt a choir, environmental interactions can cause decoherence, leading to errors in quantum computing and communication. Managing decoherence is one of the biggest challenges in building reliable quantum technologies.

Verticals of quantum technologies

Quantum computing. Quantum computing harnesses superposition and entanglement to process information in ways classical computers cannot. Instead of conventional bits, quantum computers use qubits, allowing them to perform complex calculations exponentially faster in certain cases. This has groundbreaking applications in cryptography (for example, breaking classical encryption like RSA), drug discovery (for example, simulating molecular interactions), materials science, and complex optimisation problems in logistics, finance, and artificial intelligence. While still in

its early stages, quantum computing is poised to transform industries by solving problems currently infeasible for classical supercomputers.

Quantum communication.

Quantum communication leverages the principles of quantum mechanics to enable ultra-secure data transmission. Unlike classical encryption, which relies on complex mathematical problems that classical computers cannot solve in real time (an assumption increasingly challenged by advances in quantum computing and algorithms), quantum communication ensures information-theoretic security.

Any eavesdropping attempt disturbs the quantum state, revealing the intruder's presence. The most relevant application is QKD, which is a method of securely generating and distributing random keys between parties. Beyond security, quantum networks aim to create a quantum internet, enabling distributed quantum computing and long-distance entanglement-based communication. Applications include secure government and financial transactions, defence communications, and satellite-based QKD for global cybersecurity.

Quantum sensing and metrology. Quantum sensing and metrology exploit quantum properties to achieve unprecedented measurement precision. Quantum sensors can detect minute changes in gravitational fields, magnetic fields, and time with extreme accuracy. Applications include navigation systems that function without GPS, medical imaging with improved resolution, and geological surveys for resource exploration. In metrology, quantum-based atomic clocks are revolutionising timekeeping standards essential for telecommunications, global positioning systems, and fundamental physics research.